

INDIAN INSTITUTE OF TECHNOLOGY KHARAGPUR

1. Name of the Academic Unit: Chemistry

2. Subject Name: Physical Chemistry of Solids and Soft Matter
L-T-P: 3-1-0 Credits: 4

3. Pre-requisites: NIL

4. Syllabus and reference books:

Syllabus:

Module 1: Physical chemistry of solid state:

Band theory of solids; Crystal defects and semiconductors; Qualitative aspects of physical characterization of solids: Diffraction and microscopy; Electrical, optical, and magnetic properties.

Module 2: Colloids, Surfaces and Macromolecules:

Liquids: Surface tension, viscosity, condensation; Colloidal state of matter: Properties of lyophilic and lyophobic colloids; Stability of colloidal solutions, Molecular self-assembly; Introduction to thermodynamics of electrified interfaces; Introduction to thermodynamics of surfaces: surface energy; adsorption isotherms; Macromolecules – synthetic and biological; The structures of macromolecules; Properties of macromolecules; Determination of size and shape.

Reference Books:

- 1) Atkins Physical Chemistry, 10th Edition 2014, P.W. Atkins and J. de Paula, Oxford University Press.
- 2) Physical Chemistry, 3rd Edition 2015, Thomas Engel and Philip Reid, Pearson Education.
- 3) Physical Chemistry, 7th Edition 2004, S. C. Rakshit, Sarat Book Distributors, Kolkata.
- 4) Physical Chemistry, 5th Edition 2002, I. N. Levine, Tata McGraw-Hill, New Delhi.
- 5) Physical Chemistry, 1st Edition 2015, R. Chang, Viva Books

5. Lecture-wise break-up:

Sl. No.	Topic	No. of lectures
1.	Band theory of solids	02
2.	Crystal defects and semiconductors	02
3.	Qualitative aspects of physical characterization of solids: Diffraction and microscopy	02

4.	Electrical, optical, and magnetic properties.	04
5.	Surface tension, viscosity, condensation	06
6.	Properties of lyophilic and lyophobic colloids; Stability of colloidal solutions, Molecular self-assembly	06
7.	Introduction to thermodynamics of surfaces: surface energy; adsorption isotherms	06
8.	Macromolecules – synthetic and biological; Structures of macromolecules; Properties of macromolecules; Determination of size and shape	08
Total number of hours		36